

APPLIED DATA STRUCTURES & ALGORITHMS LAB (BCS-553)

B.Tech CSE 5th SEMESTER

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Lab Assignment 10: Parent and Child Classes

Experiment No. 10.1 - Animal and Dog

Experiment Title:

Animal and Dog

Aim:

To show that a child class can use a method of its parent class.

Software / Tools Required:

Java JDK, a text editor or IDE (such as IntelliJ IDEA, Eclipse, or VS Code), and a terminal/command prompt.

Theory:

Inheritance lets a child (sub) class acquire the fields and methods of a parent (super) class using the extends keyword, which avoids rewriting code. When the parent has a constructor with parameters, the child constructor must call it with super(...) as its first statement.

Algorithm / Procedure:

1. Define class Animal with method sound().
2. Define class Dog extends Animal with an empty body.
3. Create a Dog object.
4. Call sound() on it.

Program / Queries / Commands:

```
// Name: Mankirat Singh
// ID: BCS2024542
// Roll No: 24103032571
// Assignment 10, Task 1
// Question: Create class Animal with method sound() that prints
// 'Some sound'. Create class Dog that extends Animal with no other
// code. In main, create a Dog and call sound() on it.

class Animal {
    void sound() {
        System.out.println("Some sound");
    }
}

class Dog extends Animal {
}

public class Main {
    public static void main(String[] args) {
        Dog d = new Dog();
        d.sound();
```

```
    }
}
```

Input:

No input.

Output:

```
Some sound
```

Time / Space Complexity: Time O(1); Space O(1).

Result:

The program was compiled and executed successfully. Dog used the sound() method inherited from Animal.

Viva-Voce Questions:

- What is inheritance?
- Which keyword is used for inheritance in Java?
- What are superclass and subclass?

Experiment No. 10.2 - Person and Student with super()

Experiment Title:

Person and Student with super()

Aim:

To call the parent constructor from the child constructor using super(name).

Software / Tools Required:

Java JDK, a text editor or IDE (such as IntelliJ IDEA, Eclipse, or VS Code), and a terminal/command prompt.

Theory:

Inheritance lets a child (sub) class acquire the fields and methods of a parent (super) class using the extends keyword, which avoids rewriting code. When the parent has a constructor with parameters, the child constructor must call it with super(...) as its first statement.

Algorithm / Procedure:

1. Define class Person with field name and constructor Person(String name).
2. Define class Student extends Person with constructor Student(String name) calling super(name).
3. Create new Student("Neha").
4. Print the name.

Program / Queries / Commands:

```
// Name: Mankirat Singh
// ID: BCS2024542
// Roll No: 24103032571
// Assignment 10, Task 2
// Question: Create class Person with field name, set by a
// constructor. Create class Student extending Person, whose
// constructor takes a name and calls super(name). In main, create a
// Student named Neha and print "Name:" followed by the name.

class Person {
    String name;

    Person(String name) {
        this.name = name;
    }
}

class Student extends Person {
    Student(String name) {
        super(name);
    }
}

public class Main {
    public static void main(String[] args) {
        Student s = new Student("Neha");
```

```
        System.out.println("Name: " + s.name);
    }
}
```

Input:

Student name: Neha.

Output:

```
Name: Neha
```

Time / Space Complexity: Time O(1); Space O(1).

Result:

The program was compiled and executed successfully. The Student constructor passed the name to Person using super(name).

Viva-Voce Questions:

- What does super(name) do?
- Why must super(...) be the first statement?
- Can a child class access the parent's fields?

Experiment No. 10.3 - Person and Teacher

Experiment Title:

Person and Teacher

Aim:

To show that a child class inherits both fields and methods of its parent.

Software / Tools Required:

Java JDK, a text editor or IDE (such as IntelliJ IDEA, Eclipse, or VS Code), and a terminal/command prompt.

Theory:

Inheritance lets a child (sub) class acquire the fields and methods of a parent (super) class using the extends keyword, which avoids rewriting code. When the parent has a constructor with parameters, the child constructor must call it with super(...) as its first statement.

Algorithm / Procedure:

1. Define class Person with field name and method greet().
2. Define class Teacher extends Person with an empty body.
3. Create a Teacher and set name = Kiran.
4. Call greet().

Program / Queries / Commands:

```
// Name: Mankirat Singh
// ID: BCS2024542
// Roll No: 24103032571
// Assignment 10, Task 3
// Question: Create class Person with field name and method greet()
// that prints "Hello," followed by the name. Create class Teacher
// extending Person with no other code. In main, create a Teacher,
// set its name to Kiran and call greet().

class Person {
    String name;

    void greet() {
        System.out.println("Hello, " + name);
    }
}

class Teacher extends Person {
}

public class Main {
    public static void main(String[] args) {
        Teacher t = new Teacher();
        t.name = "Kiran";
        t.greet();
    }
}
```

Input:

Teacher name: Kiran.

Output:

```
Hello, Kiran
```

Time / Space Complexity: Time O(1); Space O(1).

Result:

The program was compiled and executed successfully. Teacher used the inherited name field and greet() method.

Viva-Voce Questions:

- What is code reusability?
- Does Teacher need its own greet() method?
- What is the IS-A relationship?